CO2xNxPhrag: Porewater Methane

May 2026 Samples

2026-07-17

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##Run Information

```
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

```
## Run Information: Brenden D. Blakley
```

```
#set the run date & user name
run_date <- "06/12/2026"
sample_year <- "2026"
sample_month <- "May"
user <- "Brenden Blakley"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

GHG_files <- c("Raw Data/20260612_GHGdata_LTREB_Phrag.csv")


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/LTREB_Phrag_ShimadzuGC_QAQC_2026.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_CO2xNxPhrag_Porewater_CH4_202505.csv"

#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 25 #Equilibration Temperature (Celsius)

volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrized
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrized (should be equal to volume_g

#record any notes about the run or anything other info here:
run_notes <- ""



#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

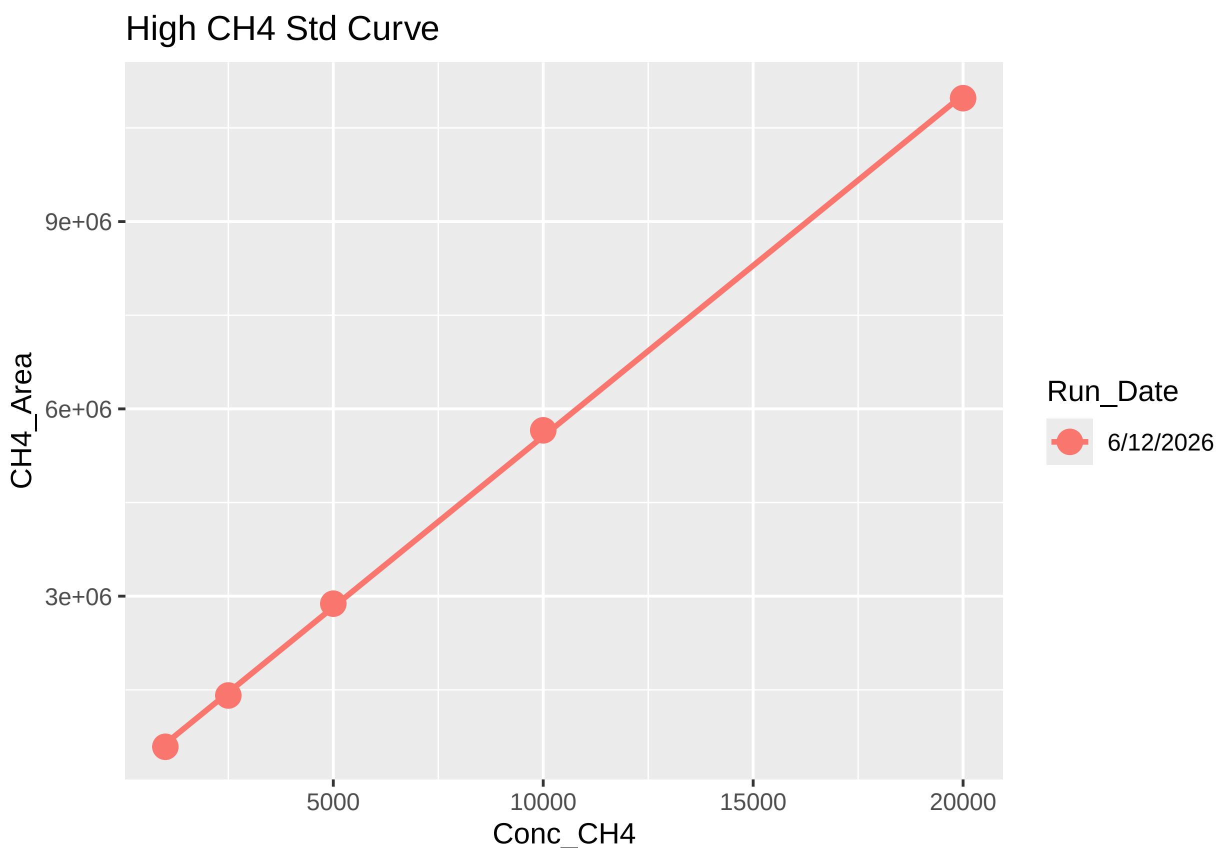
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

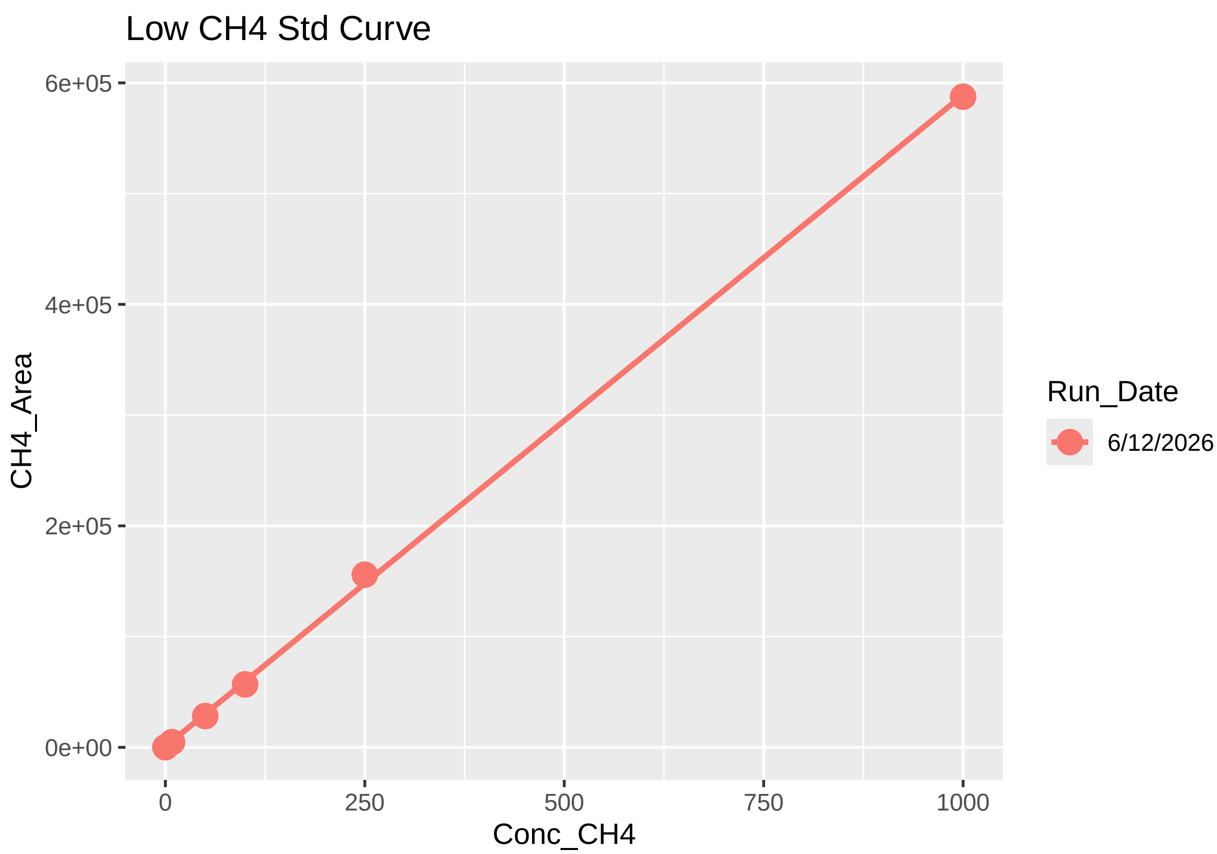
1 Read in data files

2 Assess standard curves

```
## `geom_smooth()` using formula = 'y ~ x'
```

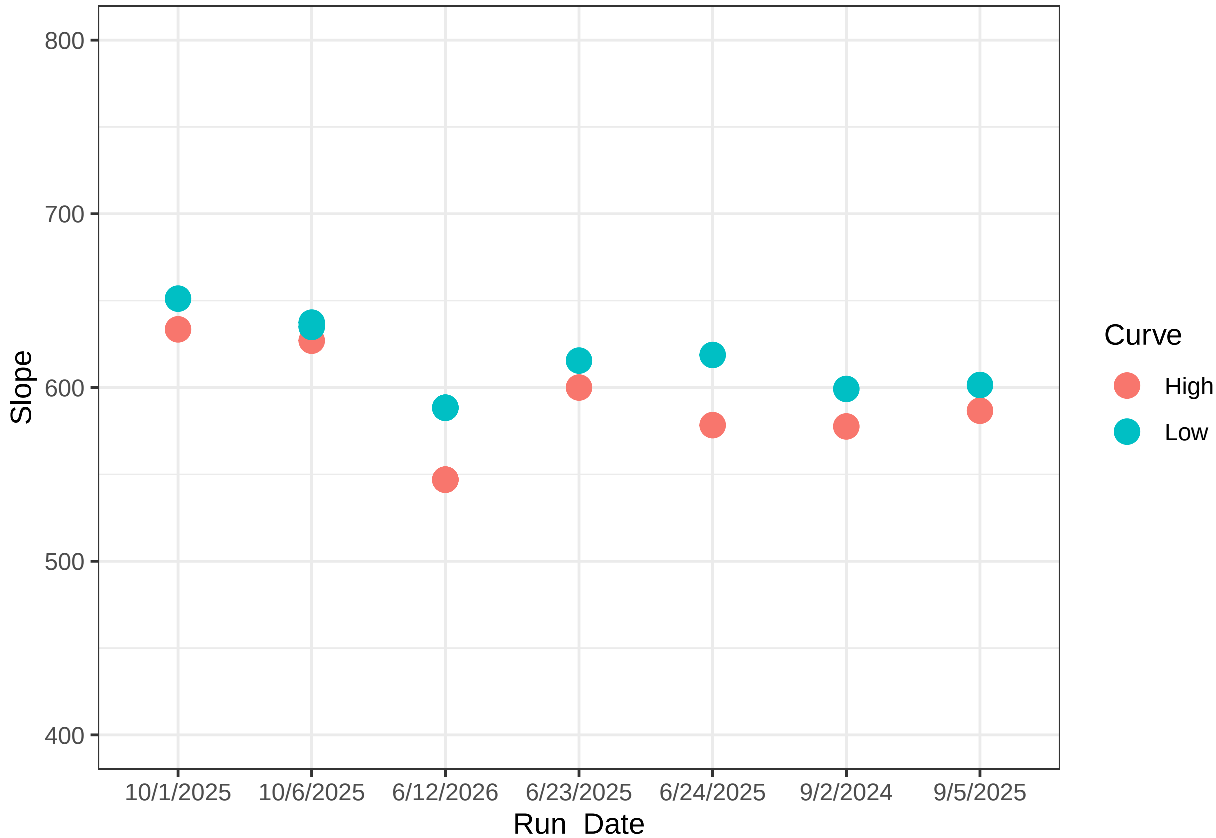



```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
## Joining with `by = join_by(Run_Date)`
```

```
##   X Curve        R2    Slope   Intercept  Run_Date
## 1 1  High 0.9997391 577.5898 104642.5944  9/2/2024
## 2 2  High 0.9998755 586.6350  48939.9343  9/5/2025
## 3 3   Low 0.9997620 599.1783    606.0063  9/2/2024
## 4 4   Low 0.9999368 601.4464    195.9643  9/5/2025
## 5 5  High 0.9999761 599.9888  21668.5842 6/23/2025
## 6 6  High 0.9991361 578.3234 147368.3056 6/24/2025
```



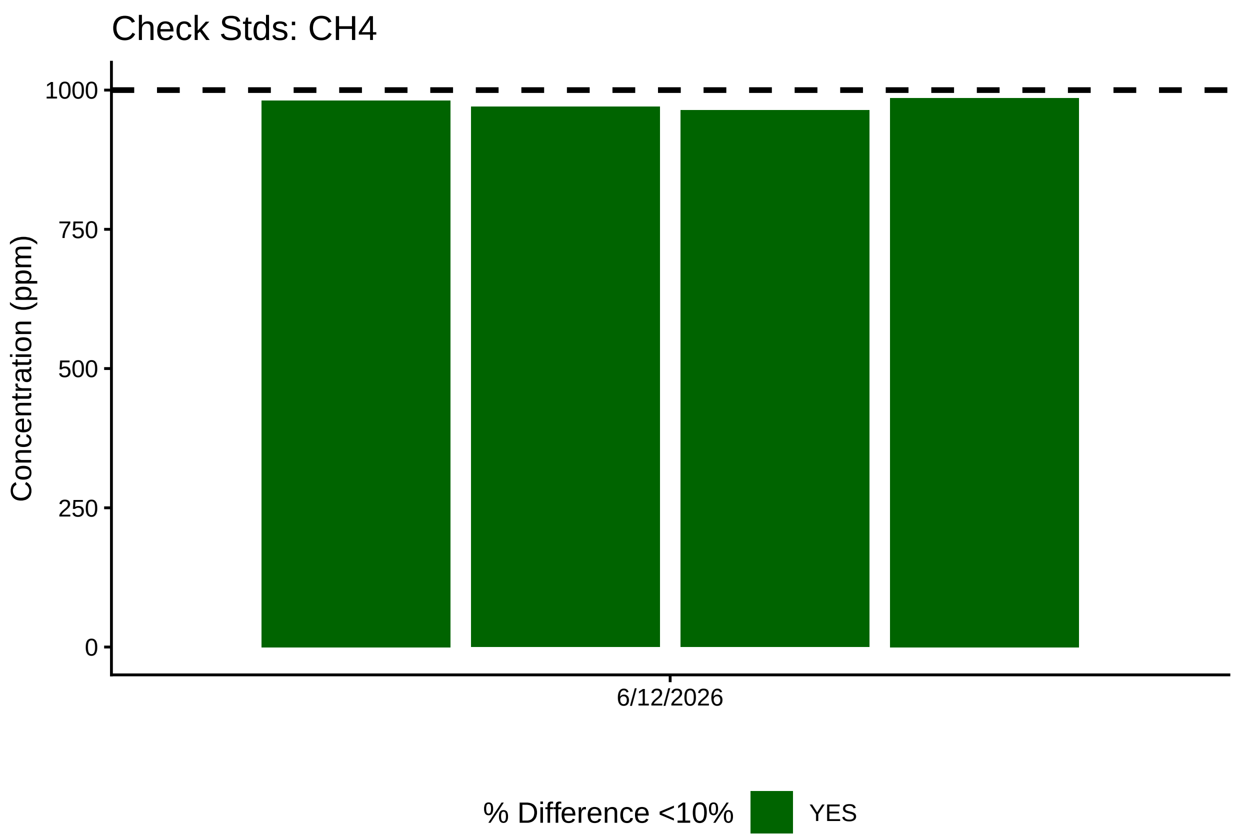

```
## [1] "High_R2 is Good-6/12/2026"
```

```
## [1] "Low_R2 is Good-6/12/2026"
```

2.1 Now calculate the CH4 & CO2 concentrations in ppm

3 Anlayze the Check Standards

```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```



4 Analyze the Blanks

```
## Assess Blanks

## >60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED
```

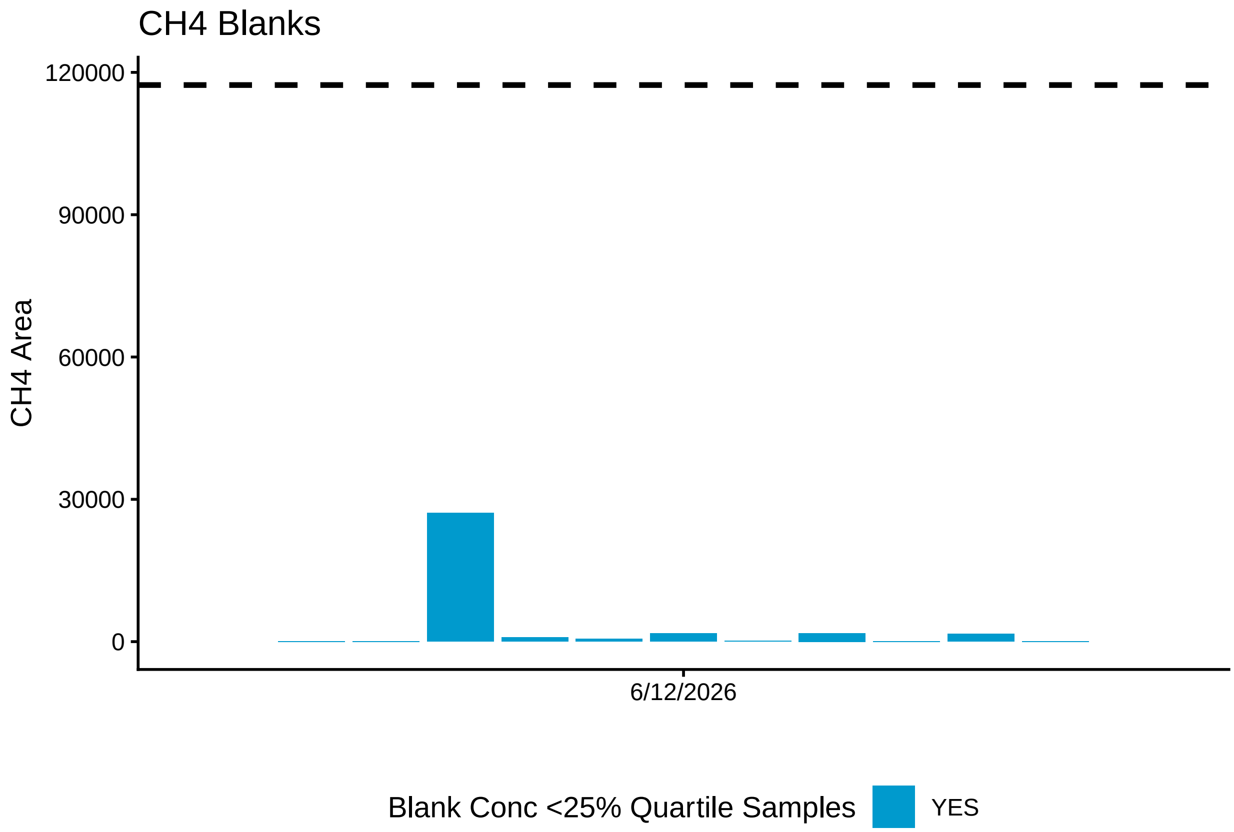
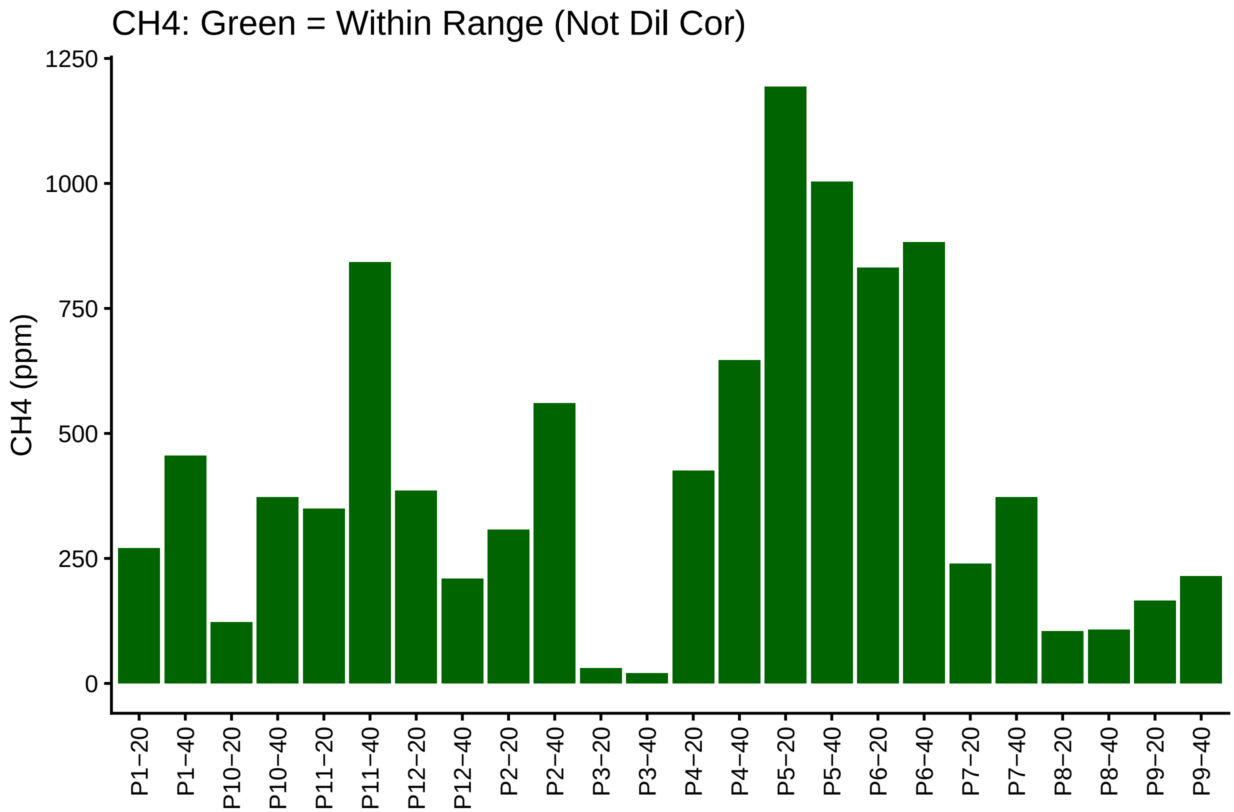
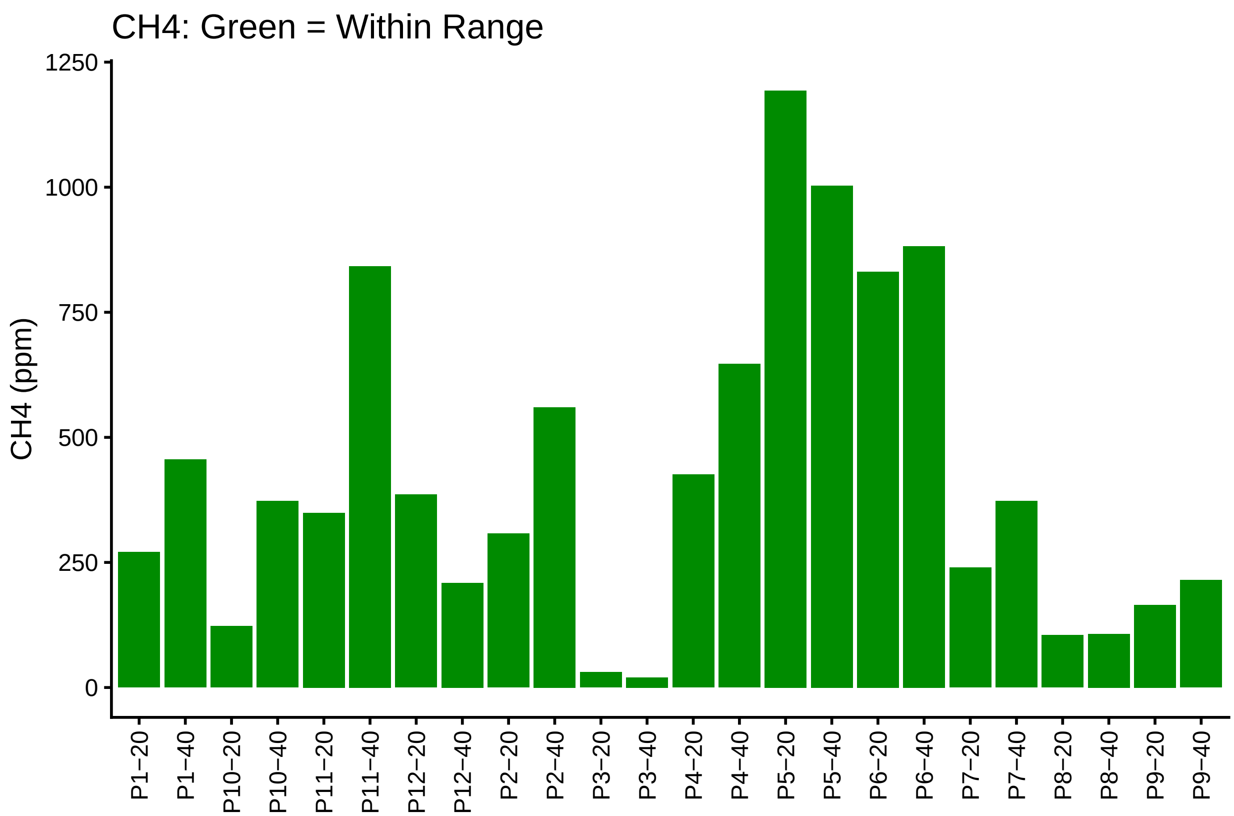


Table 1: Mean Area of Blanks

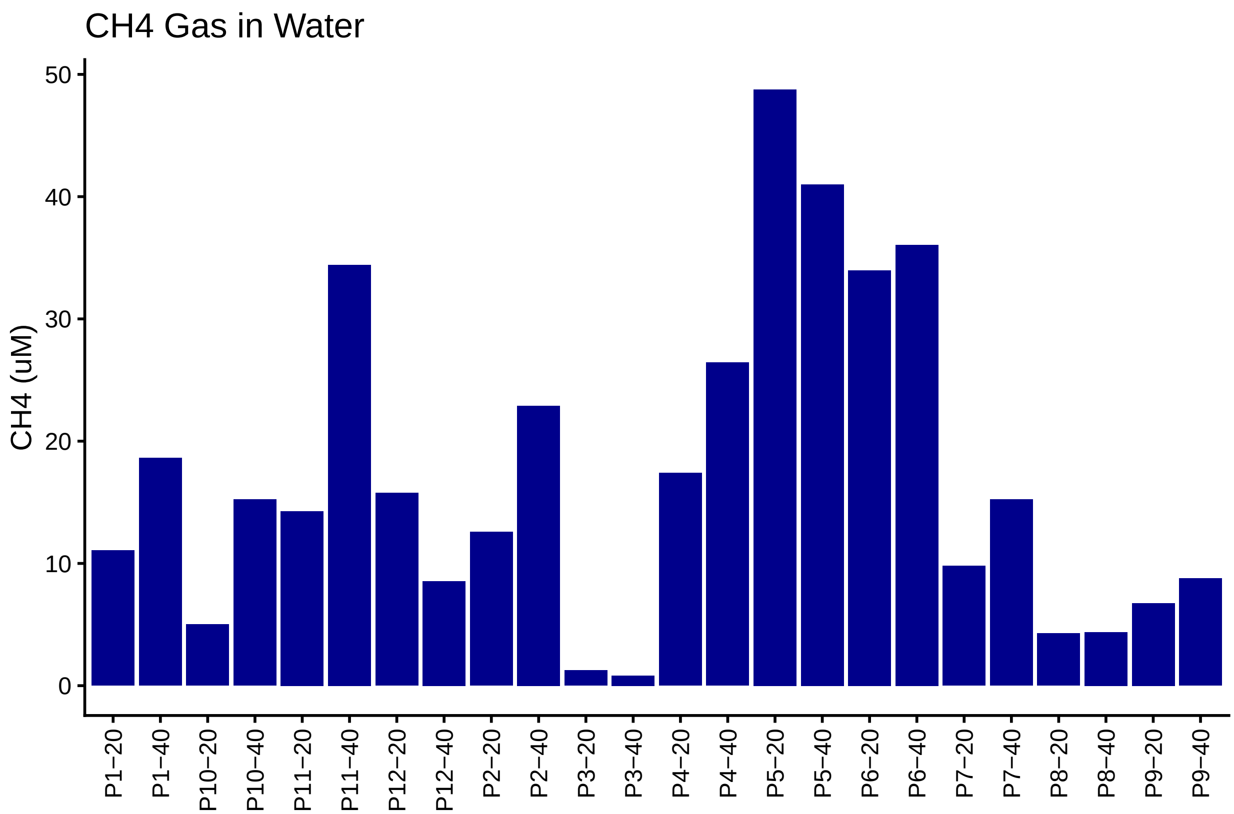
Test	Blank_Mean_Area
CH4	3125.091

5 Dilution correct samples and Check Range





6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

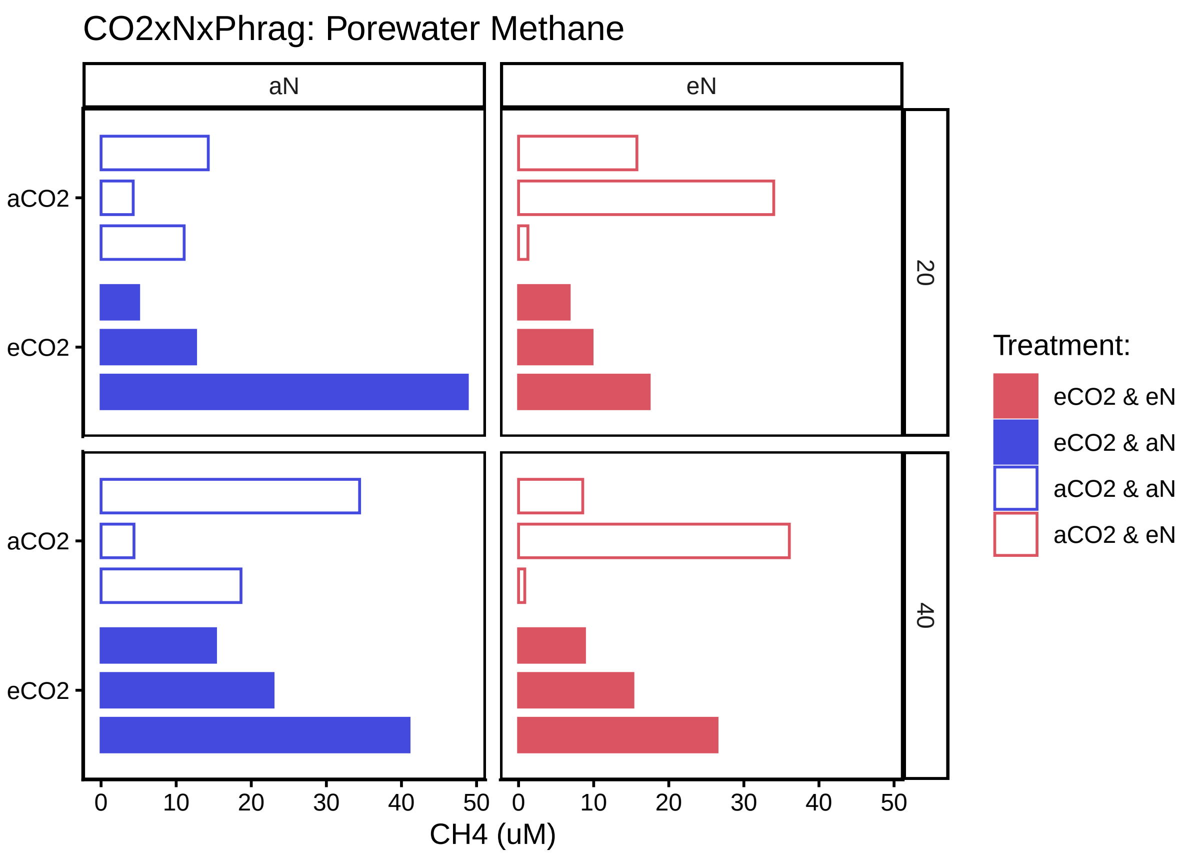
```
## Merge Samples with Metadata
```

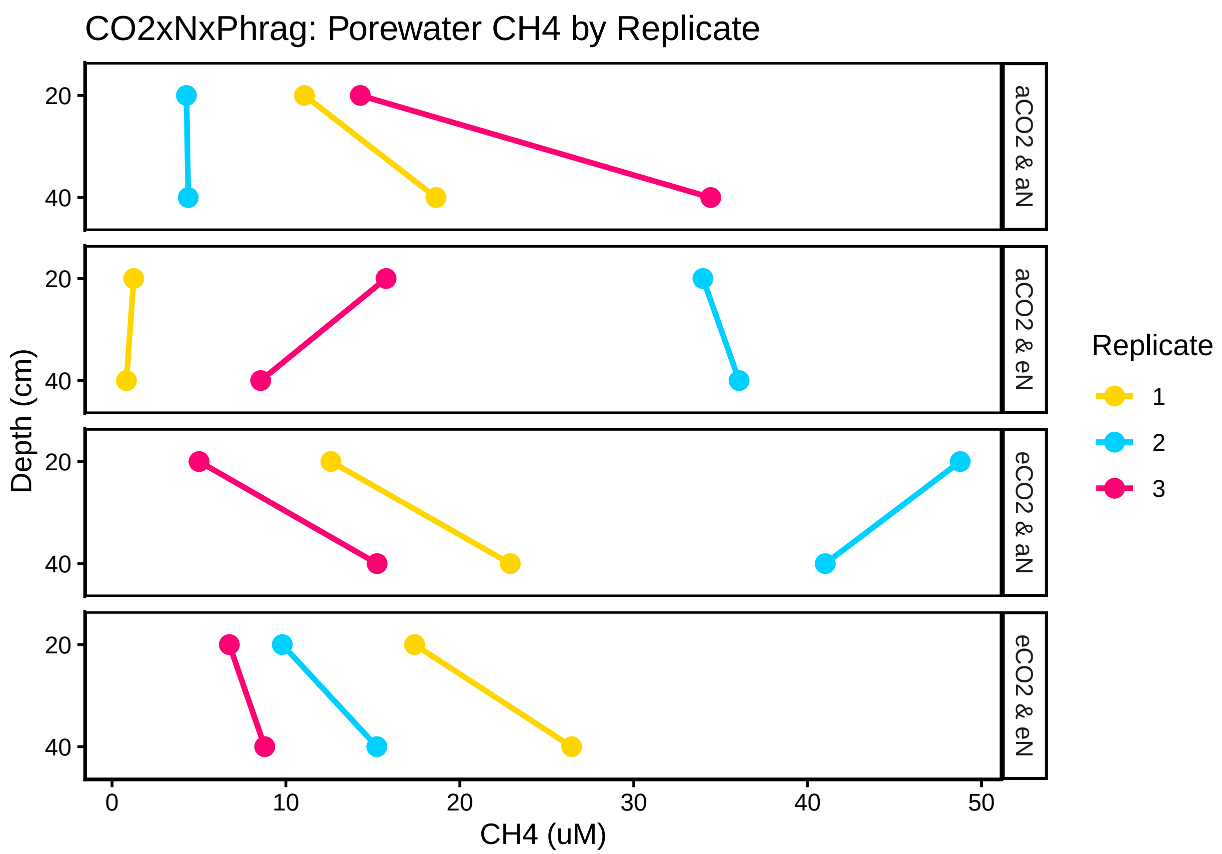
```
## All sample IDs are present in data
```

7.1 Organize Data

7.2 Visualize Data

```
## Visualize Data
```


CO2xNxPhrag: Porewater Methane
aN
eN
aCO2
eCO2
aCO2
eCO2
20
40
0
10
20
30
40
50
0
10
20
30
40
50
CH4 (uM)
Treatment:
eCO2 & eN
eCO2 & aN
aCO2 & aN
aCO2 & eN



#Export Processed Data

```
write.csv(All_Clean_Data, final_path)
```

#end